Inverse Approximation Theory for Nonlinear Recurrent Neural Networks

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Abstract

We prove an inverse approximation theorem for the approximation of nonlinear sequence-to-sequence relationships using recurrent neural networks (RNNs). This is a so-called Bernstein-type result in approximation theory, which deduces properties of a target function under the assumption that it can be effectively approximated by a hypothesis space. In particular, we show that nonlinear sequence relationships that can be stably approximated by nonlinear RNNs must have an exponential decaying memory structure - a notion that can be made precise. This extends the previously identified curse of memory in linear RNNs into the general nonlinear setting, and quantifies the essential limitations of the RNN architecture for learning sequential relationships with long-term memory. Based on the analysis, we propose a principled reparameterization method to overcome the limitations. Our theoretical results are confirmed by numerical experiments.

Recurrent neural networks

Let m be the hidden dimension. With trainable weights c, W, U and activation σ,

$$h_{k+1} = \sigma(Wh_k + Ux_k), \quad h_0 = 0 \tag{1}$$

$$y_k = c^\top h_k. \tag{2}$$

Here x are the inputs and y are the outputs.

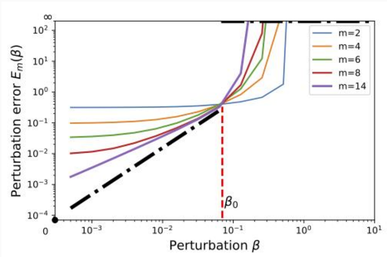
Stable approximation

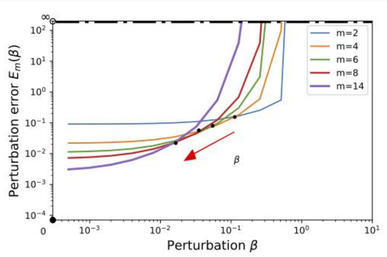
Definition 1 (Stable approximation via parameterized models)**.** Let $\beta_0 > 0$. We say a target functional sequence $\mathbf{H}$ admits a β_0-stable approximation under $\{\widehat{\mathcal{H}}^{(m)}\}$, if there exists a sequence of parameterized approximants $\widehat{\mathbf{H}}_m = \widehat{\mathbf{H}}(\cdot, \theta_m)$, $\theta_m \in \Theta_m$ which are continuous (in input $\mathbf{x}$) in point-wise topology such that

$$\lim_{m\to\infty} \|\mathbf{H} - \widehat{\mathbf{H}}_m\| \to 0, \tag{3}$$

and for all $0 \le \beta \le \beta_0$, the perturbed error satisfies that $E(\beta)$ is continuous in β for $0 \le \beta \le \beta_0$.

The following two graphs are perturbation errors for exponential decaying memory and polynomial decaying memory:





Memory function

Definition 2 (Memory function of nonlinear functional sequences)**.** For continuous, causal, regular and time-homogeneous functional sequences $\mathbf{H} = \{H_t(\mathbf{x}) : t \in \mathbb{R}\}$ on $\mathcal{X}$, define the following function as the *memory function* of $\mathbf{H}$ over bounded Heaviside input $\mathbf{u}^x$:

$$\mathcal{M}_B(\mathbf{H})(t) := \sup_{|x|_2 \le B} \frac{1}{B} \left| \frac{d}{dt} H_t(\mathbf{u}^x) \right|, \quad B > 0. \tag{4}$$

This memory function definition is a natural extension for the function ρ in linear functional.

For continuous, causal, regular and time-homogeneous functional sequences $\mathbf{H} = \{H_t(\mathbf{x}) : t \in \mathbb{R}\}$ on $\mathcal{X}$, we say it has a *decaying memory* if:

$$\lim_{t\to\infty} \mathcal{M}_B(\mathbf{H})(t) = 0, \quad \forall B > 0. \tag{5}$$

Main theorems

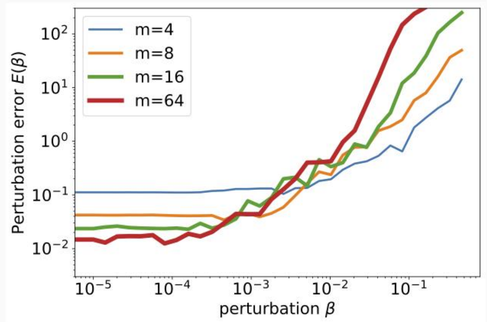
Theorem 3 (Approximation Theorem - Curse of memory in nonlinear RNNs)**.** *Assume* $\mathbf{H}$ *is a sequence of continuous, causal, regular and time-homogeneous functionals on* $\mathcal{X}$ *with decaying memory. Suppose there exists a sequence of* ***hardtanh*** *RNNs* $\widehat{\mathbf{H}} = \{\widehat{\mathbf{H}}(\cdot, \theta_m)\}_{m=1}^{\infty}$ β_0*-stably approximating* $\mathbf{H}$ *in the Sobolev norm. Then, the memory function* $\mathcal{M}_B(\mathbf{H})(t)$ *of the nonlinear functional decays exponentially for any* $\beta < \beta_0$*:*

$$\lim_{t\to\infty} e^{\beta t} \mathcal{M}_B(\mathbf{H})(t) = 0, \quad \forall B > 0. \tag{6}$$

A similar result can be proved for **tanh** activation under more assumptions.

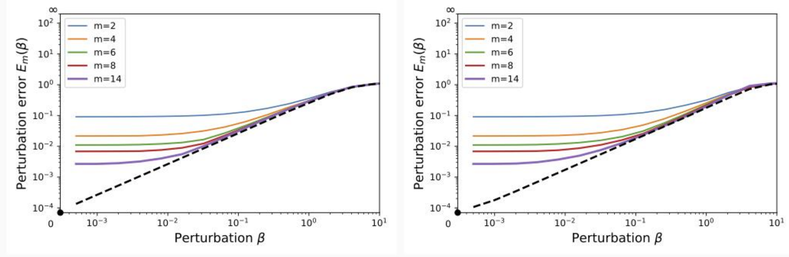
Curse of memory phenomenon in nonlinear RNNs

For **nonlinear functionals** with **polynomial decaying** memory function and **tanh** RNNs,



Reparameterization enables stable approximation

For linear RNNs, stable approximation of target functionals with polynomial decaying memory can be achieved via reparameterization of the recurrent weight matrix W. Exponential parameterization $W = -e^M$ and softplus parameterization $W = -\log(e^M + I)$ are shown:



Conclusion

We derive the first known Bernstein-type result in the setting of sequence modelling using nonlinear RNNs. We show that, assuming a given target sequence relationship (mathematically understood as a nonlinear functional sequence) can be stably approximated by RNNs with hardtanh/tanh-like activations, then the target functional sequence's memory structure must be exponentially decreasing.

References

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